

AutoEval: A NLP Approach for Automatic Test Evaluation System

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Abstract—In this modern era, there is a need for automation in answer assessment systems as the world moves towards automation. Since online response evaluation is only available for option-based questions, the evaluator's job is made more difficult when evaluating theory answers. The teacher manually checks the answer and assigns the required grade. The current system necessitates more manpower and time to evaluate the response. In this paper, we demonstrate the working and operation of the 'AutoEval - Automatic Test Evaluation System', a computerized or automatic exam paper evaluation system. A manual paper evaluation is time-consuming and exhausting, but computerized exam paper evaluation aids in standardizing the paper correction process in an effective and reliable manner. The evaluation of response scripts is an essential part of evaluating students' results. In most cases, a response evaluation is performed manually, which can lead to bias. The assessment is based on several factors including the evaluator's mood swings and the inter- relationship between the student and evaluator. Furthermore, assessment and evaluation are a time-consuming and repetitive process. Natural language processing (NLP) methods are used to evaluate exam papers in a variety of ways. NLP is used to search for grammatical errors, as well as to perform syntactic analysis, semantic similarity and database storage.

Keywords - Automatic Evaluation, Natural language Processing [NLP], NLTK, Gensim, Text Pre-processing, Similarity Measure, Marks Scoring

I. INTRODUCTION

In education systems all over the world, computers and information technology are causing a revolution [16]. They have numerous benefits, including low cost and global reach. Until now, the new generation of e-learning technologies has proven useful in a variety of fields. Online examinations are more accessible and flexible than traditional assessment methods. They minimize the overall expense of exam processing especially as they save papers, storage, and materials' costs [1]. Moreover, manually evaluating answers is a very tedious and tiresome task. Manual testing is time-consuming and involves a large amount of workforce. The paper checker may not be able to distribute marks evenly, also

[2]. Multiple choice questions, on the other hand, cannot determine a student's ability to write and express themselves. In several domains, educators prefer to use essay questions to assess students' abilities more fairly. Many researchers believe that the way assessment tools, and human graders evaluate assignments leads to great variation in grades given to students. In addition, most evaluations are performed considering specific concepts. Grades are only awarded if certain specific concepts are present; otherwise, the response is marked as incorrect. Consequently, a new system is proposed to address this problem. This paper presents a digital approach based on Natural Language Processing (NLP) for 'Automatic Test Evaluation System' to make the answer script evaluation mechanism quicker and more accurate. Natural Language Processing (NLP) is a subject of AI that works with combining human and machine languages. Speech interpretation, natural language understanding, and natural language generation are the most complicated tasks of Natural Language Processing. Computer translation, question- answering, automated text summarization, answer script evaluation, and other applications use NLP extensively. Text summarization speeds up the evaluation process by allowing precise details to be extracted from a longer text document. The similarity calculation is a technique for determining how semantically, syntactically, and structurally similar two sentences are. The scoring marks for an answer text would be determined using a similarity test. Cosine similarity, Jaccard similarity, bigram similarity, and synonym similarity are some of the algorithms used to measure similarity. Each algorithm for calculating similarity has its own meaning. Automatic test evaluation has become very popular to make the manual assessment process easier. Automatic test evaluation can be accomplished with the help of machine learning as well [3]. During the pandemic period, we, as a community, faced several tough situations. One of them is efficient and correct test evaluation of students. To assess a student's performance, a variety of evaluation methods are used. To offer an alternative, we have created an Automatic Test Evaluation System where a teacher uploads ideal answers, in accordance with the questions and students have to give their answers using a variety of grammatical styles and words which will be checked and

analyzed with the help of Natural Language Processing (NLP). The student responds to the question by writing a detailed textual answer. The test evaluation system will in return show the percentage of similarity between the uploaded answers and the ideal answers for further accuracy and proper percentage of correctness. The automatic test evaluation is inspired by the fact that it takes less time, requires fewer resources, prevents human evaluator's psychological changes, and is simple to keep track of record and extraction. The Automatic Test Evaluation based on Natural Language Processing (NLP) would assist in overcoming the challenges that manual evaluation creates. The system would automatically score points after the test as soon as the student's written response is given as input. As a result, our system will test answers based on ideal answers, saving both time and manpower. Just the student needs to put his answers as input, after which the system will compare and evaluate the answers based on the ideal answers uploaded by the teacher, and the system will assign marks depending on the similarity index of the ideal answer. This method would also reduce the evaluation error of the marks for any specific question. There is a need for such a system that can provide a simple way to evaluate answers and provide qualifying marks. This system would also assist various colleges, universities, and coaching institutes in evaluating responses in less time and with less manpower. Checking answers necessitates a high level of focus for an extended period of time, which often results in errors. The automation of this activity would improve the performance of large-scale answer evaluation. We'd have to keep track of new students, teachers, and subjects, as well as get test results from the test assessment system. It has a user interface that is both familiar and well-thought-out, as well as powerful evaluation capabilities.

II. LITERATURE REVIEW

Several approaches have been explored to solve the problems of automatic test evaluation. Some of them are summarized below:

M. Yagci and M. Unal in [4] recently released the design and development of an adaptive exam system. They propose an improved web-based adaptive exam platform for three categories of users: Administrator, Instructor, and Student, with statistical methods and assessments used to test students' data.

G. Frosini et al. in [5] define a method for designing software systems that can eliminate the task of the examiner during a standard exam, minimizing anxiety and increasing evaluation performance by computerized adaptive training. According to their model, The Database includes queries submitted by the author, the Design module handles query insertion, and the Research module analyses exam results and measures unique indexes to assess the ease and complexity of submitted queries. Desouza et al in [6] discussed the efficacy of online tests and paper exams. They also describe that student perform better on online tests. Sarayrih et al [7] discussed that an online system has many benefits, but it still has several pitfalls. They also address the challenges with online tests, including problems such as personal identities and unauthorized network intervention. Chang et al. in [8] performed a comparative analysis of the various scoring

techniques. They have focused on various kinds of exams and also how they influenced the possibility of guessing in multiple-choice questions. Mu et al. in [9] have addressed a system for automatically grading code assignments. They developed a tool and tested some of the code's problems, such as output assessment and logical errors. Sheeba Praveen in [10] states that several government and semi-government exams, such as the [IBPS Common Written Examination (CWE)], have gone online in recent years. According to the professor, we've been able to make online test arrangements for objective-type questions in recent years. As compared to an offline test, the online method contains numerous benefits. Exams that are taken online allow you to test your responses in less time. He proposed an algorithm that automatically evaluates the single-sentence descriptive answer since there were no provisions for offline paper assessment. R. Deepalakshmi et al in [11] defined OCR as the method of translating the handwritten text into a machine-readable format that can be searched, edited, and indexed. The artificial neural network is being used in the paper to achieve high precision in optically identifying the character. The proposed procedure is validated and implemented on a character dataset that holds English characters, numbers, and special characters. Another approach proposed by Baoguang et al in [12] incorporate the attributes of CNNs and RNNs (Recurrent Neural networks). In this case, the convolutional layers are built using just the convolutional and max-pooling layers from a generic CNN model (fully-connected layers are removed). CNN is able to recognize the individual characters in the answer. Conversely, since the length of inputs is static, the variable length of answers poses a challenge for CNN to define. To address this problem, RNN is combined with LSTM (Long Short-Term Memory) in order to save a long string of characters. This model was termed CRNN (Convolutional Recurrent Neural Network). In CRNN, deep features are defined as sequential representations in order to be invariant to sequence-like entity length variation. Kudi et al in [13] conducted research on the online assessment of descriptive answers using short text matching. This strategy focused on short answer matching and used a machine learning approach to solve the problem using text mining. Meena K et al in [14] researched "Evaluation of Descriptive Answers Using the Hyperspace Analog to Language Algorithm and Self-Organizing Map." These studies are based on the online assessment of the descriptive answer, which would remove the gap in the manual evaluation. To separate groups of terms, the HAL algorithm is used. Patil et al in [15] conducted a study on "Evaluating the student descriptive answer using natural language processing." The system incorporates NLP tools to analyze the paper. Authors in [17], used Manhattan- Siamese LSTM for analyzing similarity of text. The research also suggests certain marking threshold for fair evaluation of descriptive answers based on multiple surveyed parameters. We can infer from [4,5,6,7,8,9,10,11,12,13,14,15] that automated test evaluation is a prerequisite of today's educational systems and that for student assessment, it should be designed on a generic basis, as well as handle security issues and require better performance evaluation algorithms.

III. METHODOLOGY

This project is a Natural Language Processing framework for Automated Test Evaluation. The aim of this research is to automatically analyze descriptive response scripts and assign marks to each question. We use the response script as an input to accomplish this. Every algorithm is implemented using the Python programming language. The data is then processed using NLP to extract text from the response document [16][17]. Various similarity measures have been determined and are used as the criterion for grading and assigning marks. The primary goal of this newly proposed method is to evaluate the semantic significance of student responses by taking into account a variety of student responses to questions. There are several applications available, but they are all special and use different methodologies. Some applications only assess MCQs (multiple choice questions) rather than the subjective question.

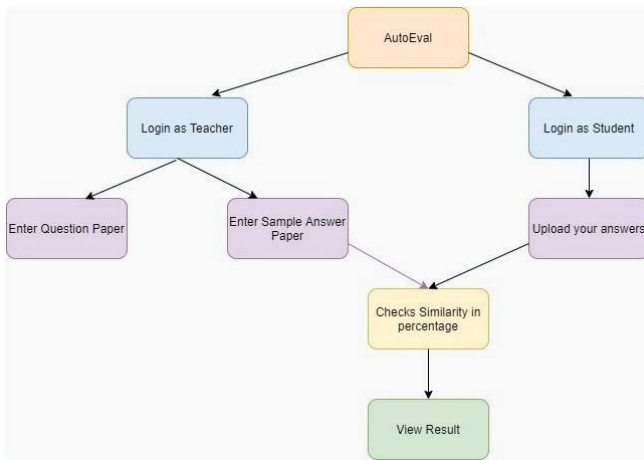


Fig. 1. Flowchart describing the algorithm

A. Input Answer

Students respond to the question by writing a thorough textual response in accordance with the questions, and they must use a variety of grammatical styles and terms in their answers, which will be checked and analyzed with the help of Natural Language Processing (NLP).

B. Text Pre-processing

Preprocessing is a method of translating data into a format that a machine can understand. Natural language processing is a powerful tool for handling text preprocessing. Tokenize text into words, delete StopWord, lemmatize word, remove redundant word, and so on are all examples of text preprocessing [18]. Natural Language Toolkit (NLTK) is a popular framework for developing Python programs that interact with human language inputs and perform NLP preprocessing. NLTK is also quite easy to learn; in fact, it is the most basic natural language processing (NLP) library available. It involves tokenization, parsing, classification/grouping, stemming, tagging and semantic reasoning text processing libraries. It has a powerful built-in feature that allows you to handle text preprocessing with less commands.

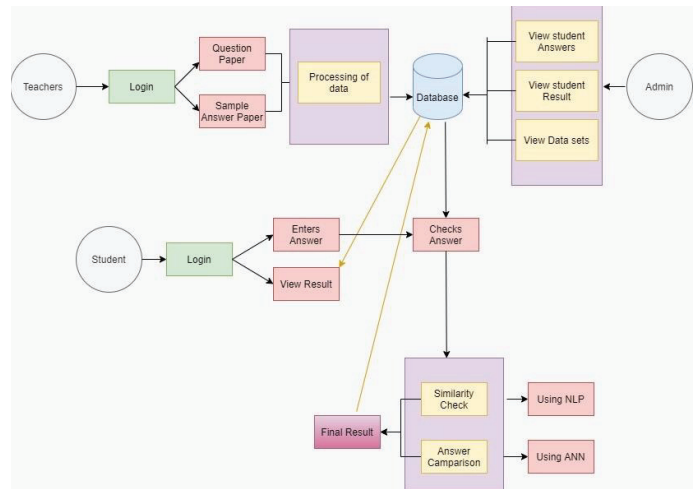


Fig. 2. Block Diagram of AutoEval

C. Tokenization

We split the paragraphs into sentences using the NLTK built in function "sent tokenization", and then use the word-tokenize method to split the sentence further into words and store them into an array. And then we count the average number of words per sentence, so we'll need to use both sentence tokenization and words to measure the ratio. Let's look at how it works:

1) *Read file and tokenize sentence*: The file will be opened and read by our software. The array would then be modified with tokenized sentences for tokenization of word.

2) *Tokenization of words and creating dictionary*: It will tokenize the words for each sentence after we add tokenized sentences to an array. We used Gensim, a Natural Language Processing package that attempts to do 'Topic Modeling for Humans', to work on text documents. However, it is considerably more than that. It is a leading and cutting-edge text processing and word vector modelling package(such as Word2Vec, FastText etc) Other packages such as scikit, R, and others, include topic models and word embedding. Gensim, on the other hand, has an unparalleled breadth and scope of operations for building and evaluating topic models, alongside a bunch of other useful text processing features. Gensim also offers the benefit of allowing you to manage huge text files without requiring the full file to be loaded into memory. The words (as known as tokens) further converted to unique ids in Gensim. As a result, Gensim allows you to construct a Dictionary object that assigns a unique id to each word. As a consequence, it will now convert the sentences to a word array and pass it onto the corpora.Dictionary() object. Each word in a dictionary is given a number. Gensim allows you to read the text and update the dictionary one line at a time without having to load the whole text file into device memory.

3) *Creating bag of words*: The Corpus (a Bag of Words) is the next important object in Gensim that we used to function. It's a simple object that stores the word id and the number of times it appears in each page (just lists the number of times each word occurs in the sentence). It is worth noting that a 'token' usually refers to a 'word'. A 'document' can be

‘sentence’ or ‘paragraph’ whereas a ‘corpus’ is a ‘bundle of documents in the form a bag of words. It will now produce a corpus of words and then it will pass the list of tokenized words to the Dictionary.doc2bow()

4) *TF-IDF*: Term Frequency – Inverse Document Frequency (TF-IDF) is a bag-of-words model that, unlike the standard corpus, down weights tokens (words) that occur frequently across documents. Tf-Idf is determined by multiplying a local component (TF) by a global component (IDF) then normalizing the result to unit length if desired [19]. The term frequency is the number of times the word appears in the text, while the inverse document frequency scales the value according to how uncommon the word is in the corpus. Simply put, terms that appear more often in the document’s lower weights.

D. Creating similarity measure object

This section presents the design of similarity object. The most important class is Similarity, which builds an index for a set of question documents provided by the teacher. The index is divided into several smaller disk-based sub-indexes by the Similarity class. The index matrix is saved in the ‘workdir’ directory, which is the same directory as the application.

E. Create Query Document

The similarity of this answer (query) document to each document in the index is explored in this section. As soon as the student types of their answer, it will include query documents or sentences, which will be tokenized as before. We obtain new documents (query documents or sentences), as a result, an existing dictionary may be updated to incorporate the new words.

F. Document similarities to query

At this point, it can be noticed that the query and all index documents have a lot in common. We ran a similarity query against the corpus to find correlations between our query document and the indexed documents. Assume that our documents consist of text: “At the party we had delicious red punch, a bunch of different appetizers, and a cookout. Since it was at the park, we played volleyball, went swimming, and sunbathed for fun” and query document is: “At the party we enjoyed food and drink and various outdoor activities”.

G. Average Similarity

Now numpy was used to measure the similarity outputs. The total of these floats was measured with the aid of Numpy. The word-similarity found was 0.267. Later, average similarity was obtained by dividing the obtained value by the number of documents. Absolute average similarity found was 0.1336, average similarity percentage was 13.36. Which means that, the main document is 13% similar to the query document. If the similarity of more than one query document. The sum of averages for each query document was estimated, yielding an overall percentage of similarity.

Let’s assume that the following as our main document: “A very difficult problem was solved by a young mathematician who lives in London. He is also a professor of Oxford University.” And our query document text is “A mathematician who is a professor found a solution to the problem.” The similarity index found was 0.3626.

IV. EXPERIMENT

When a student enters the correct information during authentication, he/she will be able to give responses and view the final result. The report then shows the final calculated scores awarded to the student answers. The student can access their results by logging in with their respective register number and their password on the result page.

The proposed system takes about 15 seconds to evaluate a response, whereas the current manual assessment takes about

60 seconds. When opposed to manual answer evaluation systems, the proposed systems save 300% of time. As compared to manual systems, the proposed system is around 75-87.5% effective. The proposed system removes all human effort and time required to determine a response. As compared to our proposed system, the amount of money spent on manual response assessment is much higher. Our proposed system is a one-time investment with minimal ongoing maintenance costs.

Screenshot of application

A. Teacher’s Login Page

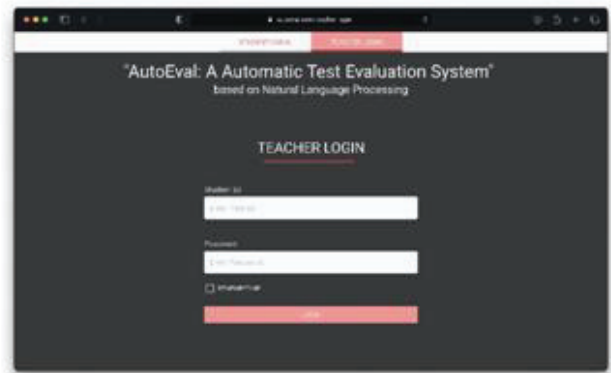


Fig. 3. Teacher Login Page

“Fig. 3” shows the teacher login page, where the teacher logs in to write the questions and the ideal responses for the specific questions. The teacher’s username id and password will be given by the college/school administration, and the teacher will be authenticated to access the creating exam page.

B. Create Exam Page (Only for Teacher)

“Fig. 4” shows the Create Exam Page, where the teacher will write the questions regarding their subjects, as well as their ideal/correct answers, which will be used to compare similarity between student’s responses and teacher’s ideal answers. All the questions will be answered in a subjective manner.

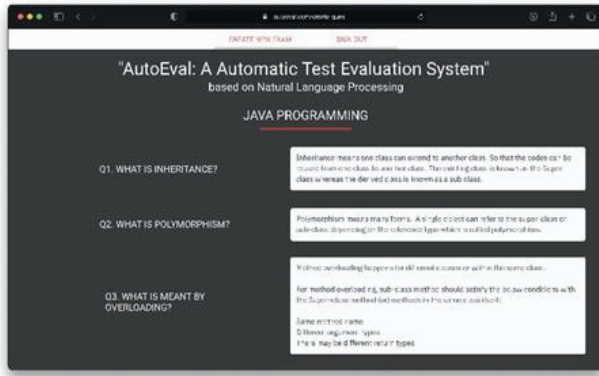


Fig. 4. Create Exam Page

C. Student Login Page

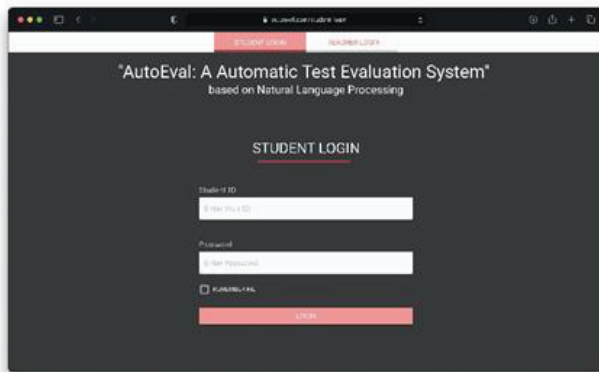


Fig. 5. Student Login Page

“Fig. 5” shows the student login page, from which the student may access the examination. The administrator will provide the login id and password, and the student will be authenticated to access the examination page.

D. Exam Page

“Fig. 6” shows the examination page, which contains questions uploaded by the teacher, which are the examination questions, which the student will answer, and then they finish their exams and see their results.

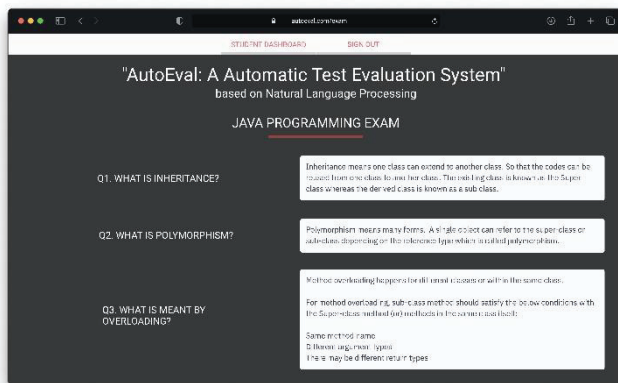


Fig. 6. Exam Page

E. Result Page

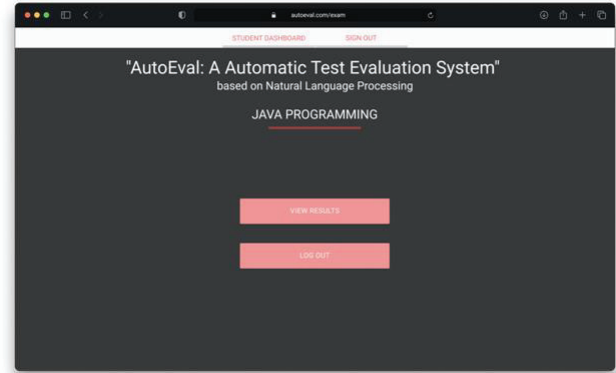


Fig. 7. Result Page

“Fig. 7” shows the student’s result page, after the student has completed his/her examination, the next page appears with the results ready to be viewed, by clicking on that, the student’s results will be shown.

V. CONCLUSION

In this research, we developed a system named ‘AutoEval: Automatic Test Evaluation System’ that uses Natural Language Processing (NLP) algorithms to calculate the results. We have developed an algorithm that will analyze theoretical answers and assign marks/grades based on the keyword matching similarity index, reducing manual work and allowing for faster evaluation. The staff will generate an answer key for the question paper for the examination process. These datasets are saved in data storage, and student responses are compared with correct answers, and the NLP algorithm is used to check for correct meaning, spelling mistakes, and grammatical errors, thus automatically calculating results using the similarity index. The length of the answer and usage of important keywords will also be considered. These algorithms have proven to be accurate and efficient. We used the NLTK library of NLP for text preprocessing, and then the word tokenization, sentence tokenization, gensim package, term frequency – inverse document frequency (tf-idf), as the resources for evaluating final marks. In future work, Deep learning approaches may be used as an innovative technology to recognize handwriting on the student response sheets. This would be more realistic and rational. We can also train our model with different handwriting using a Recursive Neural Network (RNN). The system can also be trained to analyze diagrams and assign the marks accordingly.

REFERENCES

- [1] Elsayed, Eman, Kamal Eldahshan, and Shaimaa Tawfeek. "Automatic evaluation technique for certain types of open questions in semantic learning systems." *Human-centric Computing and Information Sciences* 3.1 (2013): 1-15.
- [2] Sinha, Prince, Ayush Kaul, Sharad Bharadia, and Sheetal Rathi. "Answer evaluation using machine learning." (2018).
- [3] Rahman, M., and Fazlul Hasan Siddiqui. "Nlp-based automatic answer script evaluation." vol 4 (2018): 35-42.
- [4] M. Yagci and M. Unal, Designing and implementing an adaptive online examination system, *Procedia - Social and Behavioral Sciences*, 116 (2014) 3079-3083.

- [5] G. Frosini, B. Lazzerini, F. Marcelloni, *Performing automatic exams, Computers & Education*, 1 June 1998.
- [6] E. Desouza, M. Fleming, A Comparison of In-Class and Online Quizzes on Student Exam Performance, *Journal of Computing in Higher Education*, Vol. 14(2), pp. 121-134.
- [7] M.A. Sarrayrih, M. Ilyas, Challenges of Online Exam, Performances and Problems for Online University Exam, *IJCSI International Journal of Computer Science Issues*, Vol. 10, Issue 1, No 1.
- [8] Chang S-H, Lin P-C, Lin Z-C International forum of educational technology & society (IFETS). Measures of partial knowledge and unexpected responses in multiple-choice tests 2007, 95–109.
- [9] Mu L, Qian X, Zhang Z, Zhao G, Xu Y International conference on computer science and software engineering, vol 5. In An assessment tool for assembly language programming. IEEE Xplore Digital Library, Wuhan, Hubei; 2008:882–884. doi:10.1109/CSSE.2008.111 doi:10.1109/CSSE.2008.111
- [10] Sheeba Praveen, “An Approach to Evaluate Subjective Questions for Online Examination System”, Assistant Professor, Dept. CSE, Integral University, Lucknow, U.P, India.
- [11] B Vanni, M. Shyni, and R. Deepalakshmi, ”High accuracy optical character recognition algorithms using learning array of ANN” in Proc. 2014 IEEE International Conference on Circuit, Power and Computing Technologies (ICCPCT), 2014 International Conference.
- [12] Baoguang Shi, Xiang Bai and Cong Yao, “ An End-to-End Trainable Neural Network for Image-based Sequence Recognition and Its Application to Scene Text Recognition ”, arXiv:1507.05717v1 [cs.CV] 21 Jul 2015.
- [13] Pooja Kudi and Amitkumar Manekar (2014), “Online Examination with short text matching”, IEEE Global Conference on Wireless Computing and Networking.
- [14] Meena k and Lawrence Raj (2014), “Evaluation of the Descriptive answers using the Hyperspace Analog to language and self-organizing Map”, IEEE International Conference on Computational Intelligence and Computing Research.
- [15] Shweta M. Patil and Prof. Ms. Sonal Patil, “Evaluating the student descriptive answer using natural language processing” International Journal of Engineering Research and Technology, Volume 3, Issue 3.
- [16] Bahel, Vedant, Preeti Bajaj, and A. Thomas. "Knowledge discovery in educational databases in indian educational system: A case study of ghrce, nagpur." In *2019 International Conference on Computational Intelligence and Knowledge Economy (ICCIKE)*, pp. 235-239. IEEE, 2019.
- [17] Malsa N., Vyas V., Gautam J., Shaw R.N., Ghosh A. (2021) Framework and Smart Contract for Blockchain Enabled Certificate Verification System Using Robotics. In: Bianchini M., Simic M., Ghosh A., Shaw R.N. (eds) *Machine Learning for Robotics Applications*. Studies in Computational Intelligence, vol 960. Springer, Singapore. https://doi.org/10.1007/978-981-16-0598-7_10
- [18] Tajammul M., Shaw R.N., Ghosh A., Parveen R. (2021) Error Detection Algorithm for Cloud Outsourced Big Data. In: Bansal J.C., Fung L.C.C., Simic M., Ghosh A. (eds) *Advances in Applications of Data-Driven Computing*. Advances in Intelligent Systems and Computing, vol 1319. Springer, Singapore. https://doi.org/10.1007/978-981-33-6919-1_8
- [19] Bahel, Vedant, and Achamma Thomas. "Text similarity analysis for evaluation of descriptive answers." *arXiv preprint arXiv:2105.02935* (2021).